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Brief Communication

Predictability and Coping With Separation in Infant Squirrel Monkeys

T. C. Jordan, Christopher L. Coe, Jeff Patterson, and Seymour Levine
Department of Psychiatry and Behavioral Sciences
Stanford University School of Medicine

Twelve infant squirrel monkeys were separated from their mothers once every 48 hr for a total of 20 separations. Half of the infants experienced separations for a fixed interval (FI) of 30 min. The remaining 6 infants experienced separations for a variable interval (VI) which averaged 30 min. During the separation periods, movement and vocalizations were recorded. Blood was sampled for later cortisol assay after the 1st, 5th, 10th, 15th, and final separation. The FI group showed increased levels of movement and vocalization over repeated separations when compared with the VI group. Cortisol showed significant increases over basal levels and remained elevated throughout the entire period in both groups. These findings suggest that predictability may be more stressful than unpredictability when organisms cannot control the predicted event.

It is well established that separation of an infant from its mother during early life is a relatively traumatic experience (Bowlby, 1969, 1973; Hinde & Spencer-Booth, 1970). The severity of the effect has been shown to vary with time and under differing social and environmental constraints (Coe & Levine, 1981; Mineka & Suomi, 1978). Typically, the measures used to evaluate the degree of disturbance have been behavioral, either movement as a reflection of increased agitation or vocalizations (Rosenblum & Plimpton, 1981). Recently, physiological indexes such as heart rate and body temperature (Reite, Short, Kaufman, Stynes, & Pauley, 1978) and plasma cortisol levels (Coe & Levine, 1981; Mendoza, Smotherman, Miner, Kaplan, & Levine, 1978) have also been employed in an effort to evaluate the internal state of the infant.

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Requests for reprints should be sent to Christopher L. Coe, Department of Psychiatry and Behavioral Sciences, Stanford University School of Medicine, Stanford, California 94305.

Relatively less attention has been paid to factors that may help alleviate or mitigate the distress resulting from separation, although there are some notable exceptions (Kaufman & Rosenblum, 1967; Robertson & Robertson, 1971). Two of the most important aspects of the situation that may modify the response to stressful events are control and predictability. Numerous studies have indicated that an organism that has either control or predictability during a stress-provoking situation is generally more capable of coping effectively with the stress (Gliner, 1972; Seligman, 1975; Weiss, 1970). It has been proposed that a significant component of the infant's development involves the learning of the contingencies of prediction and control (Levine, 1980; Lewis & Goldberg, 1969; Watson, 1967; for review, see Gunnar, 1980); for example, proximity and contact with the mother are achieved either by direct intervention or by causing the mother to seek proximity through a complex set of signals. Here the infant is able to exercise considerable control, and the mother's responses are usually quite predictable. One of the most stressful aspects of involuntary separation, therefore, may be the loss of either or both of these factors, because the separated infant is no longer capable of utilizing its behavioral responses to cope in its customary manner.

A variety of previous studies have indicated that predictable stressors are less noxious than unpredictable stressors (Badia, Harsh, & Abbott, 1979; Hymowitz, 1979). However, such observations have generally been based on proce-

dures involving electric shock. It is unclear whether the same processes would be operative in the separation paradigm. Despite the extensive research on separation, few studies have systematically evaluated the effects of repeated separation and reunion on either children or animals. The previous studies have tended to use relatively few separations, often spaced quite widely apart, such that each separation might almost be considered as a new experience, and contingencies would correspondingly be less likely to be learned; or they have examined the prolonged response to extremely traumatic separation regimens which may result in chronic emotional disturbance (Mineka, Suomi, & DeLizio, 1981; Suomi, Mineka, & DeLizio, 1983). We therefore decided to investigate the hypothesis that response to separation may be modified by the degree of predictability in the situation and examined the infant's reaction to repeated brief separations. Specifically, we evaluated whether repeated separations for a fixed interval (FI) of time would lead to a differential response relative to similar separations for a variable time interval (VI). To the extent that the infants could learn the contingencies, we anticipated that FI separations would provide greater predictability than the VI situation and that, as a consequence, the responses to separation under this condition would adapt more rapidly.

Method

Subjects

Twelve mother-infant pairs of squirrel monkeys (*Saimiri sciureus*) of the Guyanese variety were used in this study. The infants were between 6 and 7 months old at the beginning of testing. They were housed in two social groups (six dyads in each) in a controlled environment with constant temperature and humidity and were on a natural light cycle. Three mother–infant pairs were drawn from each group for each condition.

Procedures

All infants were separated from their mothers once every 48 hr over a 6-week period (20 separations in all). The infants in one group were separated for a fixed interval of 30 min on each occasion and placed in individual cages within auditory, but not visual, contact of one another. Infants in the second group were also separated on alternate days except that their separations were of a variable duration with a mean of 30 min (range, 6–60). Separation times of 30 min were scheduled to occur for both groups on Days 1, 5, 10, and 15 in order to allow equivalent comparisons during these sessions. The final separation period (Day 20) was extended to 60 min for both groups in order to examine the effects of a "violation of expectancy" for the FI group. Blood samples were taken at the end of Sessions 1, 5, 10, 15, and 20 for later cortisol assay. Animals were anesthetized with ethyl ether, and 0.5 ml of blood was collected in heparinized syringes by cardiac puncture. Blood samples for each animal were obtained within 3 min of capture, and samples were collected from all subjects within 10 min. (Further details of sampling procedures and radioimmunoassay for cortisol have been presented by Coe et al., 1978.) In order to determine basal cortisol levels, blood samples had been taken 10 days prior to any of the above manipulations.

Agitated movements and vocalizations were recorded during all of the separations. These observations were conducted through a one-way viewing window from an adjoining room. One lateral movement was counted each time the subject's full body moved across the vertical midline of the cage. Two observers recorded these movements, and reliability tests indicated that this measure could be easily scored in a standardized manner ($r = .96$). Initially, vocalizations of each infant were also scored by direct observation; however, it was quickly felt that this approach was unreliable, as the animals moved rapidly and often faced in the opposite direction from the observers. Consequently, total vocalizations for the entire group were recorded on a Model 4000 Uher tape recorder from which a complete tally could be accurately determined. The tape-recorded vocalizations were scored by two independent observers (blind with respect to group and day), and the mean value was taken. This scoring system proved extremely reliable (Spearman-Brown $r = .99$).

Results

Movement

The numbers of movements in the first 30 min of each session were totaled for all sessions in which the VI group had been separated for at least 30 min. These scores could thus be compared with those of the FI infants that were always separated for exactly 30 min. Figure 1 illustrates these data at the standardized 5-day intervals when blood samples were taken.

Analysis of variance (ANOVA) compared conditions (VI and FI) and separation sessions

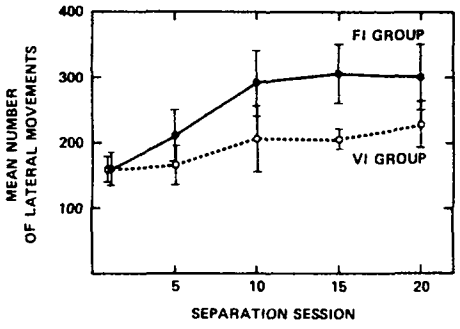


Figure 1. Mean (±*SD*) number of movements per session for the infants separated at fixed (FI) and variable intervals (VI).

(days) which were treated as a repeated measures factor. The ANOVA indicated a highly significant main effect of repeated separations, $F(11, 110) = 10.04$, $p < .001$, and there was a significant interaction with the type of separation, $F(11, 110) = 2.17$, $p < .02$. As suggested by the figure, both groups showed high levels of agitated movement as a function of repeated separations, and this effect was enhanced for the infants experiencing the fixed-interval separations. The effect emerged rapidly between Day 1 and Day 10 of separation; thereafter, the groups remained relatively constant in activity levels in relation to each other. Both groups still showed high levels of activity on the final day when the separation was extended to 60 min to test the reaction of the FI infants. This extension did not differentially affect the FI group; infants from both treatment conditions showed the same significant decline in movement from the first 30 min to the second 30 min of the test, $F(1, 10) = 10.8$, $p < .01$.

Vocalization

Vocalization rates, as already noted, were based on group scores between Day 8 and Day 20. The daily scores were relatively consistent over this period, and the mean levels of vocalizations for the FI and VI subjects are portrayed in Figure 2A. A two-tailed independent *t* test of five equivalent sessions for each group showed that the difference in vocalization level was significant, $t(8) = 4.04$, $p < .01$. When totals were broken down into 6-min blocks (see Figure 2B), it can be seen that the differential effect of the separation conditions persisted throughout the session, although both groups showed a marked decline in calling within a session. The final test, which involved extending the separation to 60 min, did not differentially affect this decline. Both groups showed an additional 250% decline in calling during the second 30-min period, although the decrease was somewhat greater for the VI group.

Cortisol

Plasma cortisol levels showed significant elevations over basal values on all days (1, 5, 10, and 20), $F(5, 50) = 7.8$, $p < .01$, and there was no indication of a reduction in the cortisol response in either the VI or the FI animals. The adrenal response appeared to be greater in the VI infants; however, this difference was already apparent on Separation Day 1 before any differential manipulations had taken place, and thus we cannot attribute it specifically to any selective effect of unpredictability in this experiment.

Discussion

The results of the experiment confirm previous studies that have shown that separation of infant squirrel monkeys from their mothers

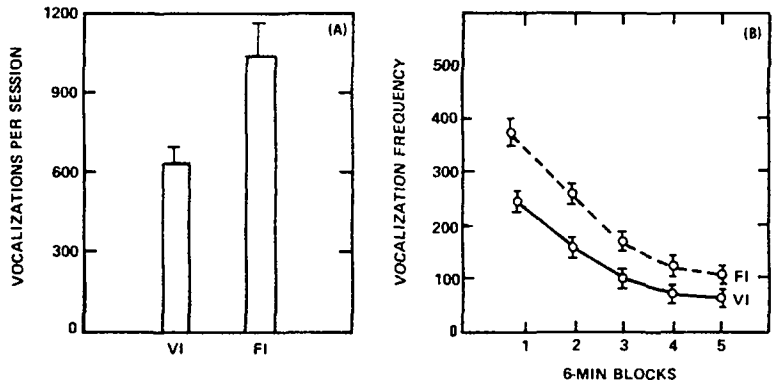


Figure 2 A: Mean (±*SD*) number of vocalizations for fixed- (FI) and variable-interval (VI) groups across days. B: Mean number of vocalizations at 6-min intervals within the 30-min separation period.

results in a marked activation of the pituitary-adrenal system (Coe & Levine, 1981; Mendoza et al., 1978) and more recently that this activation does not decline over several separations (Coe, Glass, Wiener, & Levine, 1983). The significant cortisol elevations over basal levels attested to the continued stressful nature of the separation experience in both groups. Behavioral assessment also indicated that the infants did not adapt to the separation experience as a function of frequent, repeated exposures; indeed, the data suggested that separation may become more aversive as a function of time.

The infants' vocal responses to separation remained high, and although they showed within-sessions decrements, they did not decline as one might have anticipated from previous studies on rhesus macaques, in which the frequency of the infants' vocalizations decreased both within *and* across sessions (Suomi et al., 1983). Coe et al. (1983) reported data on separated squirrel monkey infants in which the decline in calling bore a striking resemblance to an extinction curve. There were a number of procedural differences that may account for this difference. Infants in the current experiment were separated as a group, and although they were visually screened from each other, they were in auditory contact. It is quite possible, therefore, that calling levels were maintained at relatively high rates by one infant's response facilitating the continued response of other infants. It may be that it is only when in complete isolation that calling declines rapidly across sessions, where presumably this is due to the ineffectiveness of calling in eliciting the normal reinforcer, the mother. In addition, the separations in the current work occurred at 2-day intervals, which may evoke a sensitized response in contrast to the behavioral adaptation observed in the infants that are separated at 10-day intervals (Coe et al., 1983).

Whereas the vocal and movement responses of the VI group remained relatively constant, the agitated behavior of the FI infants actually increased across separations. An affective interpretation would lead one to state simply that the infants were more emotionally disturbed by the FI separations. However, it is also possible to entertain a more cognitive interpretation based on coping theory. In this view, the vocalizations would reflect signaling attempts on the part of the infant to elicit maternal proximity-seeking and caregiving. Thus, the capacity of the FI separated infants to predict imminent reunion with the mother might have provided greater incentive for calling. Moreover, prior experience with reunion at a predictable time point might have enhanced the infants' sense that calling was efficacious in causing reunion. This functional explanation must be verified, however, because the final separation, which tested the "violation of expectancy" hypothesis, failed to evoke higher calling levels in either FI or VI infants.

It should also be noted that these behavioral findings differ significantly from the worsening behavioral response during prolonged separations reported by other investigators (Mineka et al., 1981; Suomi et al., 1983). Those studies on rhesus monkey infants were concerned primarily with the onset and intensity of the inactive/depressed response to separation, not the initial "protest" behavior. We observed no freezing, body rocking, or self-clutching in any of our subjects. We do not find this surprising, inasmuch as the inactive/depressed syndrome has characteristically been induced by more prolonged and traumatic separations than were utilized in the present study. The depressive response is also more typically observed in infants reared without mothers (Kraemer & McKinney, 1979), when additional demands are imposed on the separated infant (Kaufman & Stynes, 1978) or when the reunion environment is also disturbed (McGinnis, 1980).

The movement and vocalization data also appear to be in conflict with other results concerning the general effects of predictability, as most investigators report that unpredictability is more aversive than predictability (for reviews, see Badia et al., 1979; Hymowitz, 1979; MacDonald, 1973). In accounting for the apparent discrepancy, it is important to consider the nature of the stressor and the animal's ability to cope with the aversive situation. Weinberg and Levine (1980) noted that many studies that found predictability to be beneficial utilized paradigms that provided the animal with some measure of control over the stressor, such as a choice of conditions (predictable or unpredictable) under which shock is administered (e.g., Gliner, 1972; Lockard, 1963) or postural control over the painfulness of the shock (Biederman & Furedy, 1973). In contrast, paradigms that have not offered any element of control have often found predictability to be more stressful than unpredictability (e.g., Brady, Thornton, & Fischer, 1962; Pare, 1964). The separation paradigm may fall into the latter category, because reunion is completely under the control of the experimenter. In this case, the increased behavioral disturbance evinced by the FI group would be accounted for by the fact that an aversive situation is made more stressful by predictability in the absence of control.

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